

actually investing in a physical prototype; the virtual model provided unlimited information about the product and its design at various operational scenarios. “Over a period of time, the application of digital twins increased, driven largely by the amount of data that gets generated through instrumented assets. The insights generated through digital twins increase tremendously whenever they can be applied in real-time and invariably brings in an element of predictability in addressing run-time failures,” says Sampath Kumar Venkataswamy, Senior Research Manager for Manufacturing Insights at IDC Asia/Pacific.

Digital twins have gained popularity in India over the past two to three years in spaces such as manufacturing, utilities, and oil and gas, though for most companies it is at an R&D or proof-of-concept (PoC) stage. “In the next three to five years as these PoCs mature, we visualise large-scale adoption of digital twins, targeted towards specific business needs of an organisation,” says a Wipro spokesperson.

Industries such as oil and gas, minerals, metals and mining, power, renewables, pulp and paper, and pharma are focussing on easier optimisation of products and processes by using digital replicas. “For instance, a typical oil refinery can incur losses to the tune of millions due to unscheduled shutdowns and recommissioning efforts. By investing and implementing in a digital twin, these organisations can pre-empt shutdowns and create a targeted ramping down plan to keep production disruptions to a minimum,” says Venkataswamy. Its application can be across the value chain, with organisations looking to understand the supply chain impact and

its effect on manufacturing operations, which in turn would affect customer deliverables.

There are several examples of companies using digital twins. Cairn Oil and Gas was one of the first upstream companies to take steps to implement a comprehensive digital operations programme. In India, this private oil and gas explorer deployed Honeywell Forge enterprise performance management (EPM) software to improve efficiency, while enabling workers to remotely operate its facility. Telecom, too, is a big adopter, with Finnish network equipment maker

PERIENCE platform of Dassault Systèmes to develop and deliver innovative, high-quality trucks and buses to the growing commercial vehicles market in the country, in a cost-effective manner. Similarly, Gurugram-headquartered JBM Group—which is into automotive, buses and electric vehicles, among others—uses Dassault’s virtual twin to develop next-gen products, from conceptualising design to manufacturing and even extending to after-sales support. Deepak Thakur, CEO of JBM (Bus Division), says, “Using this technology, we were able to map the entire new product development lifecycle on a single digital thread.”

This technology is also good for the environment. “Carbon digital twins are part of an emerging segment for most companies in India. They can help in visualising the reduction of the carbon footprint of equipment and processes by analysing data,” says Kap Prabhakaran, VP of Engineering at Honeywell Connected Enterprise.

The cost factor

There is no standard cost when it comes to building a digital twin as it varies by company and the sophistication of the technology developed. The cost is largely from the infrastructure needed to generate, store and process the data. Some other key components include digital twin software, IoT sensors, integrated platforms, compute solutions, cloud & infrastructure, and training. Rajesh Gharpure, EVP & Global Head of Manufacturing, Energy & Industry 4.0 at Larsen & Toubro Infotech (LTI) explains: “The initial setup cost for a digital twin is high because of the technology components required to implement a basic digital twin use case



Nokia introducing its 5G Digital Design concept to simulate 5G use cases. Using machine learning algorithms, the platform can leverage this technology to monitor and assess the impact of 5G implementation, while providing automated recommendations. “The solution has benefitted several of our CSP [communications service provider] customers across the globe,” says a Nokia spokesperson.

While an increasing number of automotive companies have adopted digital twins, VE Commercial Vehicles, a joint venture in India between the Volvo Group and Eicher Motors, adopted the 3DEX-